Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What Statistics Is For

You already know how to count things and record what you find — how many bottles are in the crate, how many pencils are in the tin. Statistics is the way we turn that kind of counting and recording into something useful.

Statistics is the branch of mathematics that deals with collecting, organising, and interpreting information. It is not just about counting for the sake of counting. The whole point is to take information that is scattered around — a list of names, a pile of counters, a row of marks — and arrange it so that it tells you something clear.

Think about it in three parts:

1. Collecting — you gather the information, like counting how many of your classmates are present.
2. Organising — you arrange that information in a neat way, such as putting the numbers in a table or a list.
3. Interpreting — you read the organised information to understand what it means, like seeing which day had the most pupils in school.

So the purpose of statistics is simple: it helps us collect, organise, and make sense of information. Whenever you have a lot of facts or numbers and you want to understand them, statistics is the tool you use to do it.

Practice

1. In your own words, what is statistics, and what is its purpose?
2. Name the three things statistics helps us do with information.
3. A teacher counts the number of pupils present in each class and writes the results in a table so she can see which class has the most pupils. Which two parts of statistics is she doing?

Section 2: Making Sense of a Frequency Table

Statistics is the branch of mathematics that helps us gather information, organise it, and make sense of it. When we want to answer a question like "Which colour do most students in my class like best?" or "How many children are in each family?", we first have to collect the data — that is, gather the facts or numbers. We can collect data by counting, by asking people (a questionnaire), by watching, or by recording what we observe.

Once the data is collected, it usually looks like a long, messy list. To understand it, we need to organise it. One of the best ways is to build a frequency table. A frequency table shows us each different value and how many times it appears. That "how many times" is called the frequency.

Here is the simple idea behind a frequency table: we take every piece of data, and for each value we make a mark. We often use tally marks — four vertical strokes, with the fifth drawn across them as a diagonal line, so we can count quickly in groups of five. After we have marked every piece of data, we count the marks for each value and write the total. That total is the frequency.

When the table is finished, we can interpret it — we can read the table to answer questions about the data, such as which value is the most common, which is the least common, or how many pieces of data there are altogether (the total of all the frequencies).

Worked example

A teacher asked each child in a class to say how many pets they have. Here are the answers from the class:

2, 1, 3, 2, 1, 0, 2, 3, 1, 2, 0, 2, 1, 3, 2

Step 1 — List the different values.
The answers we saw are 0, 1, 2 and 3. Write these in a column in order (smallest first): 0, 1, 2, 3.

Step 2 — Tally the data.
Go through the list one number at a time and put a tally mark next to that value. For the first five numbers, "2, 1, 3, 2, 1":

0: no mark yet
1: || (two marks — the 1s in positions 2 and 5)
2: || (two marks — the 2s in positions 1 and 4)
3: | (one mark — the 3 in position 3)

Now continue through the rest of the list — "0, 2, 3, 1, 2, 0, 2, 1, 3, 2" — adding a mark each time. When you finish, count the marks for each value:

0 appears in positions 6 and 11 → 2 marks
1 appears in positions 2, 5, 9, 13 → 4 marks
2 appears in positions 1, 4, 7, 10, 12, 15 → 6 marks
3 appears in positions 3, 8, 14 → 3 marks

Write these as tally marks in a second column and the counts as a third column called Frequency:

<table>
<tr><th>Number of pets</th><th>Tally</th><th>Frequency</th></tr>
<tr><td>0</td><td>||</td><td>2</td></tr>
<tr><td>1</td><td>||||</td><td>4</td></tr>
<tr><td>2</td><td>~~~|</td><td>6</td></tr>
<tr><td>3</td><td>|||</td><td>3</td></tr>
</table>

Check your tally.
The frequencies must add up to the total number of pieces of data. Here, 2 + 4 + 6 + 3 = 15, and there are exactly 15 numbers in the list — so the table is correct. If the frequencies did not add up to 15, we would know to re-tally.

Interpreting the table.
Now we can read answers straight from the table:

The most common number of pets is 2, because it has the highest frequency (6).
The least common number of pets is 0, because it has the lowest frequency (2).
The total number of children asked is 2 + 4 + 6 + 3 = 15.

Practice

1. Prepare a frequency table for the following data, which shows the number of children in each of 12 families:

   1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2

   First tally the data, then write the frequencies in a table with columns for "Number of children", "Tally" and "Frequency". Check that your frequencies add up to the total number of families.
2. A class counted the number of books each student read last term and recorded these results:

   3, 1, 2, 3, 4, 2, 1, 3, 2, 4, 2, 3

   Use the data to answer these questions without re-counting the whole list:
   (a) What is the highest number of books read by any student?
   (b) What is the lowest number of books read?
   (c) Which number of books was read most often?
   (d) How many students are in the class?

Check your work

Attempt every question before you read this page.

What Statistics Is For

1. Statistics is the branch of mathematics that deals with collecting, organising, and interpreting information. Its purpose is to help us collect, organise, and make sense of information.
2. Collecting, organising, and interpreting information.
3. She is collecting the information (counting the pupils) and organising it (putting the results in a table) so she can interpret it.

Making Sense of a Frequency Table

1. There are 13 families in the list. Frequencies: 1 → 3, 2 → 3, 3 → 4, 4 → 3. Total = 3 + 3 + 4 + 3 = 13 ✓

   The data has four different values: 1, 2, 3 and 4. Tally each number:

   - 1 appears in positions 1, 2, 12 → 3 marks → frequency 3
   - 2 appears in positions 7, 8, 13 → 3 marks → frequency 3
   - 3 appears in positions 3, 6, 9, 10 → 4 marks → frequency 4
   - 4 appears in positions 4, 5, 11 → 3 marks → frequency 3

   Check: 3 + 3 + 4 + 3 = 13, and the list has 13 numbers, so the tally is correct.

   <table>
   <tr><th>Number of children</th><th>Tally</th><th>Frequency</th></tr>
   <tr><td>1</td><td>|||</td><td>3</td></tr>
   <tr><td>2</td><td>|||</td><td>3</td></tr>
   <tr><td>3</td><td>||||</td><td>4</td></tr>
   <tr><td>4</td><td>|||</td><td>3</td></tr>
   </table>
2. (a) 4 books (b) 1 book (c) 3 books (it occurs 4 times) (d) 12 students

   (a) The largest value in the list is 4, so the highest number of books read is 4.
   (b) The smallest value in the list is 1, so the lowest number of books read is 1.
   (c) Count how many times each number appears:
   - 1 appears in positions 2 and 7 → twice
   - 2 appears in positions 3, 5, 9, 11 → 4 times
   - 3 appears in positions 1, 4, 8, 12 → 4 times
   - 4 appears in positions 6 and 10 → twice

   The most common is 3 (tied with 2 at 4 times).
   (d) The number of students equals the total of the data: there are 12 numbers in the list, so 12 students.

   (Note: the "most often" answer reads from the frequency table — both 2 and 3 occur 4 times, the highest frequency.)

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-702072db	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____